Wavelet Selection

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Key Concepts

1) Many different wavelets and families of wavelets are available.

2) There is no universally optimal wavelet.

3) A general goal for selecting a wavelet is to minimize the number of nonzero terms in the basis expansion. This usually results in good compression and filtering properties.

4) The number of vanishing moments and support affect the efficiency of orthogonal wavelet families that satisfy the multiresolution property.

a) A wavelet has p vanishing moments if $\int t^m \psi(t) dt = 0$ for $m = 0, 1, 2, \ldots, p-1$.

b) The support of a wavelet is the time interval over which it is nonzero.

5) The coefficient of a wavelet with p vanishing moments is zero if the signal is described by a p-term Taylor series over the support of the wavelet.

6) The support of a wavelet increases as the number of vanishing moments increase. A wavelet with p vanishing moments has minimum support $2p - 1$.

7) The coefficient of a wavelet that spans a "singularity", or time at which the character of the signal changes, is generally nonzero.

    a) Choose p large if the singularities are isolated. This results in zero coefficients for complex (polynomial order $p-1$) signals.

    b) Choose p small if the singularities are closely spaced. This reduces the number of wavelets that span singularities.

8) Daubechies' wavelets achieve minimum support for a given number of vanishing moments.

Video Screenshots

Selecting a wavelet - many options

MATLAB: Daubechies (45), Coiflets (5), Symlets (44), ...

not examine each family

Big picture

- Vanishing moments (p) $\int_{-\infty}^{\infty} t^{\ell} \psi(t) dt = 0, \; 0 \le \ell < p$
- Support (wavelet duration)

Goal: many zero coefficients

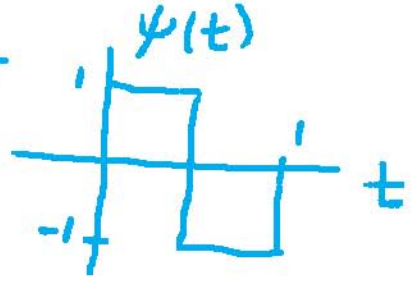
Recall $d_j[n] = \int x(t) \psi_{j,n}(t) \, dt$

Vanishing moments -

Smooth $x(t) \approx \alpha + \beta t + \gamma t^2 + \delta t^3 + \cdots$

$\psi(t)$: p vanishing moments $\Rightarrow d_j[n] = 0$ if $x(t)$ p-term Taylor series over support of $\psi_{j,n}(t)$

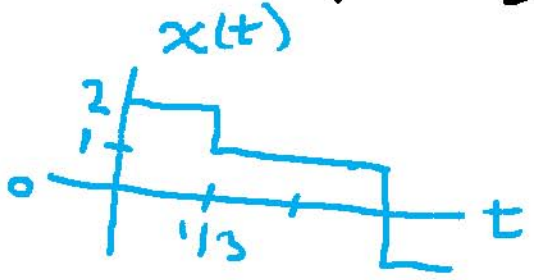
Example: Haar



$$\int_{-\infty}^{\infty} t^0 \, \psi(t)\, dt = 0 \Rightarrow p = 1$$

$d_j[n] = 0$ for $x(t)$ locally constant

$\psi_{j,n}(t)$ overlap singularities $\Rightarrow d_j[n] \neq 0$



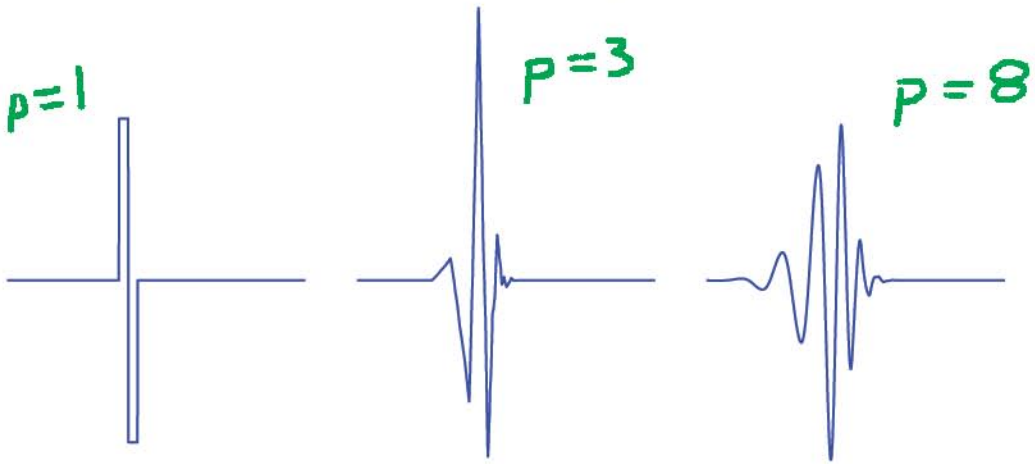
$$\int x(t)\,\psi(t)\,dt = 2\left(\tfrac{1}{3}\right) + 1\left(\tfrac{1}{6}\right) - 1\left(\tfrac{1}{2}\right) = \tfrac{1}{3}$$

Want $\psi(t)$ short duration (small support)

Orthogonal wavelets: $\psi(t)$ p vanishing moments $\Rightarrow$

$$\text{support}\{\psi(t)\} \geq 2p-1$$

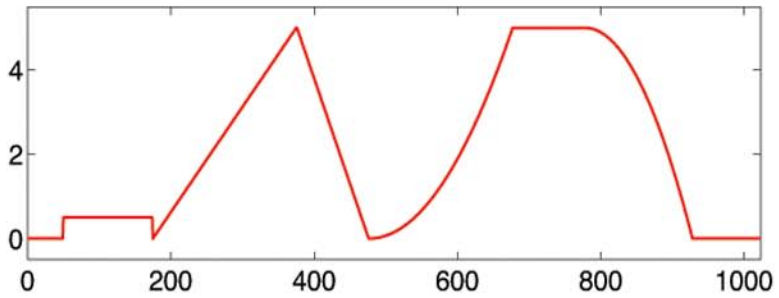
Daubechies wavelets: minimum support for given p



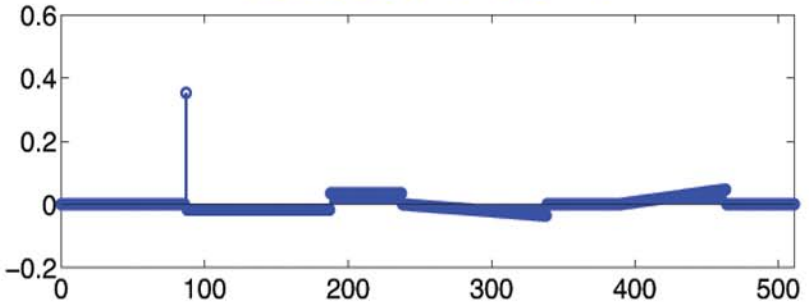
Isolated singularities $\longrightarrow$ p large

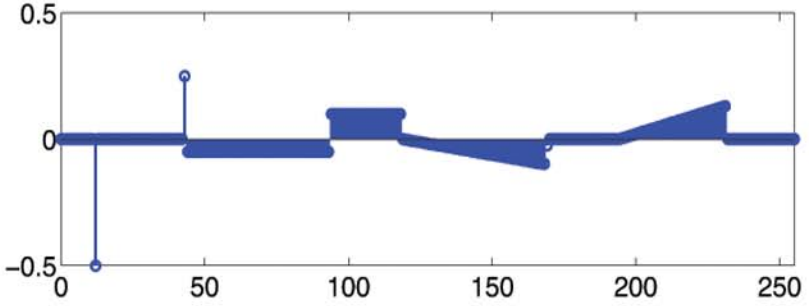
Dense singularities $\longrightarrow$ p small

Example: $p=1$ vanishing moment: Haar

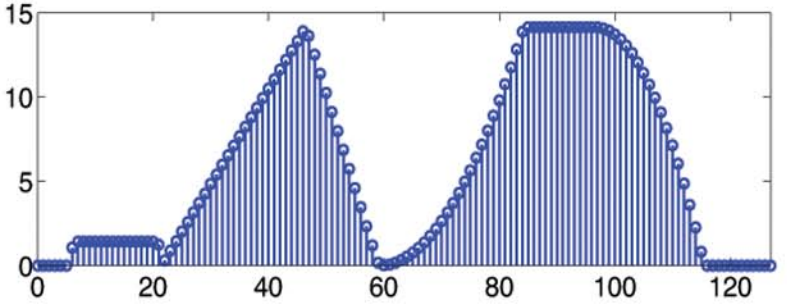


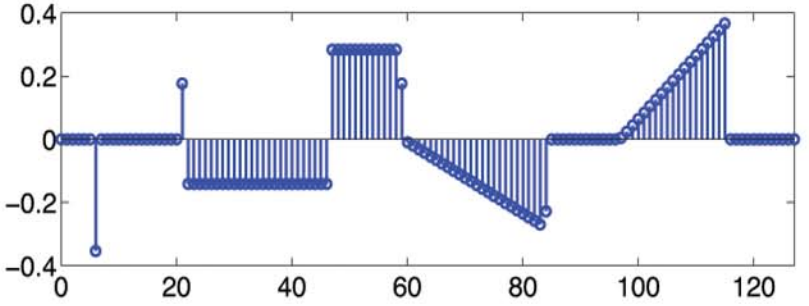
wavelet coeffs



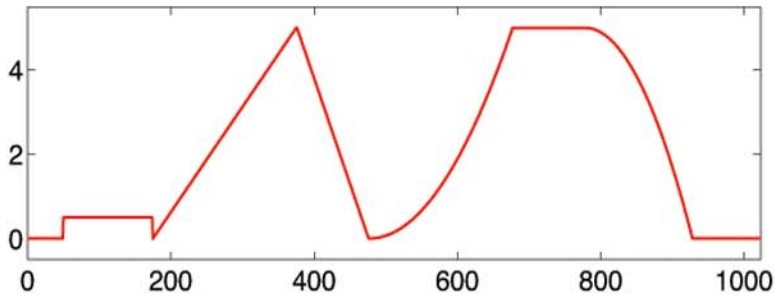


scaling coeffs

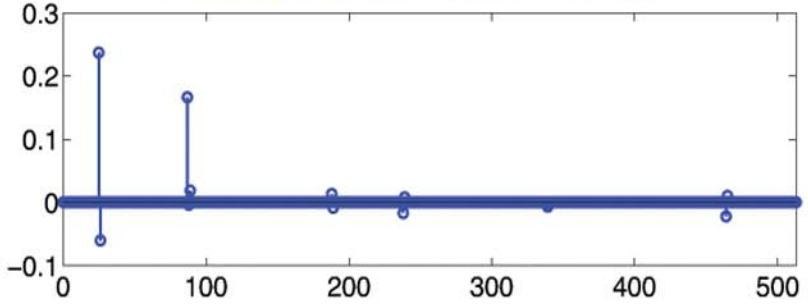


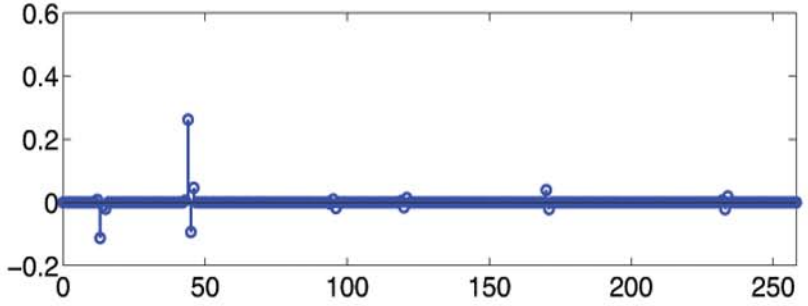


Example: p=3 vanishing moments, Daubechies 3

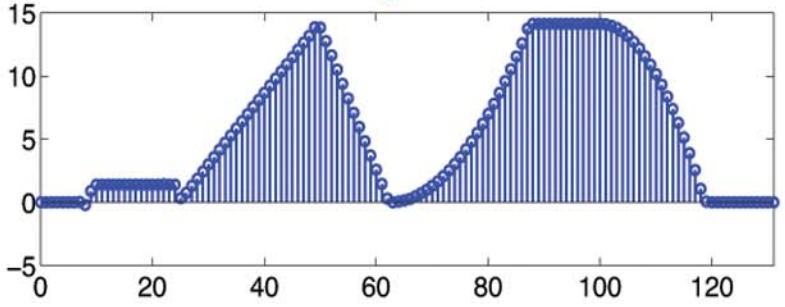


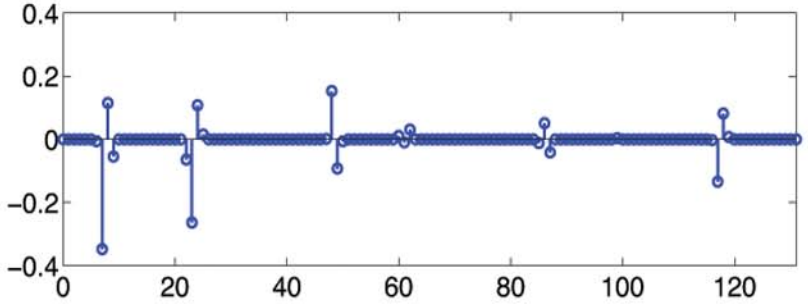
wavelet coeffs



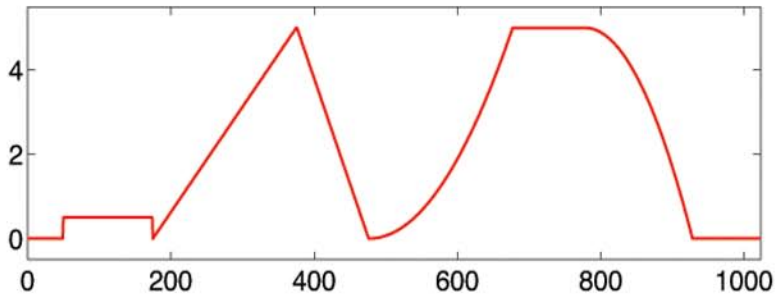


scaling coeffs

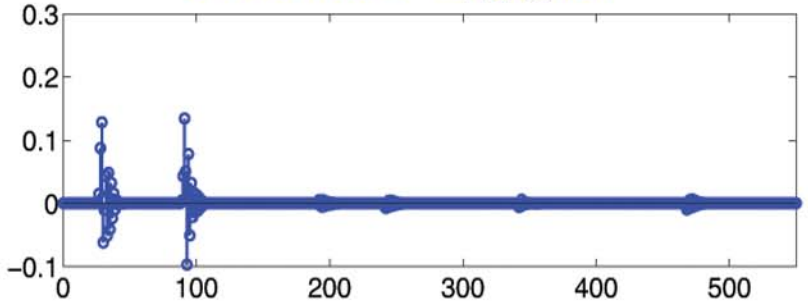


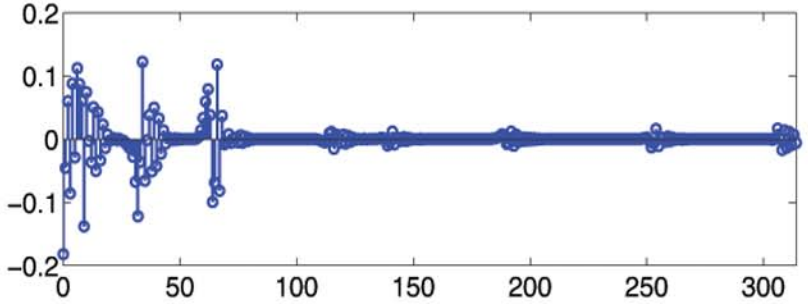


Example: p = 40 vanishing moments, Daubechies 40

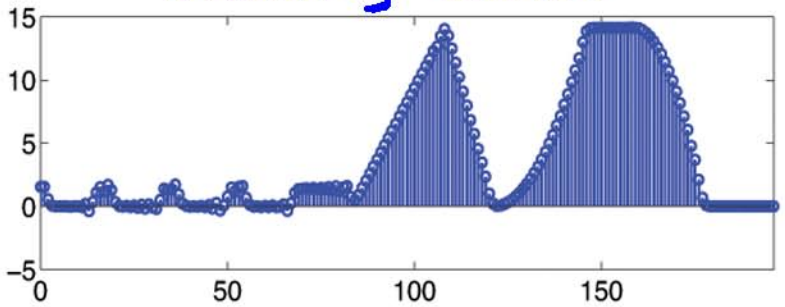


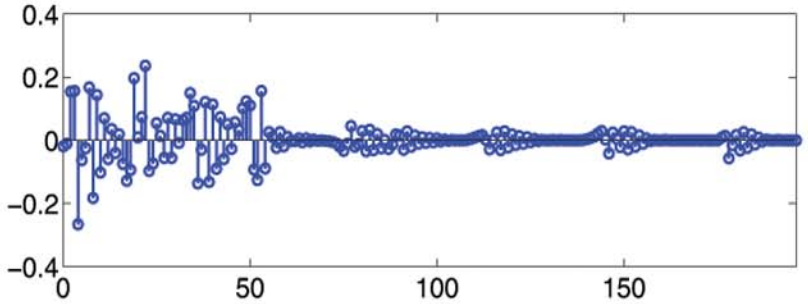
wavelet coeffs





scaling coeffs





Daubechies: p vanishing moments, support $2p-1$

Symlets: more symmetric than Daubechies

p vanishing moments, support $2p-1$

Coiflets: $\int \phi(t)\,dt = 1$, $\int t^k \phi(t)\,dt = 0 \quad 1 \le k < p$

$\int t^k \psi(t)\,dt = 0 \quad k < p$

Support $3p-1$

improved sampling accuracy

$\vdots$

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